

SalesGenie AI – AI Sales Assistant & Lead Intelligence Platform

Team Members & Roles:

No.	Team Member	Role
1	Ritika Kushwaha	Project Lead / Scrum Master, Backend Developer
2	Aditi Bhosure	Database / DevOps Engineer
3	Samraiyam Nallagorla	AI/ML Engineer
4	Tejashwini M	Frontend Developer
5	Rushikesh Arepalli	QA / Test Engineer

Project Overview:

SalesGenie AI is an AI-based sales assistant designed to help sales teams manage their leads and better understand potential customers. It brings different sales-related activities together in one platform, such as **lead management, lead analysis, lead scoring, AI-powered outreach, recommendations, CRM integration, and sales analytics.**

The main purpose of the project is to make the sales process easier and reduce the amount of manual work involved in handling leads. Instead of manually going through every lead and deciding what to do next, SalesGenie AI uses AI to analyze the available information and provide useful insights and recommendations.

The application is being developed in a modular way, with different features working together to support the overall sales process, from **managing a lead to analyzing it and deciding the next appropriate action.**

Project Objective:

The main objective of **SalesGenie AI** is to create an intelligent sales assistant that makes it easier for sales teams to manage and work with their leads. The platform is designed to help sales teams:

- Manage and organize leads in one place.
- Store and access prospect information easily.
- Analyze lead and company details using AI.

- Generate lead qualification scores to understand the potential of each lead.
- Create personalized outreach messages based on lead information.
- Suggest suitable follow-up actions for sales teams.
- Support CRM integration and conversation intelligence in future stages.
- Provide useful sales insights through a centralized platform.

Overall, the project aims to make the sales process **simpler, faster, and more organized** by combining regular lead management with AI-based analysis and assistance.

Project Modules:

The **SalesGenie AI** project is divided into six main modules. Each module focuses on a different part of the sales process, and together they form the complete platform.

Module 1 – Lead Management & Prospect Database

This module is used to manage all the leads and prospects in one place. It allows users to **add, view, update, search, and delete leads**, as well as import lead information when needed.

Module 2 – Lead Intelligence & Company Analysis

This module uses AI to understand the available **lead and company information**. It analyzes the data and provides useful insights that can help the sales team understand the potential of a lead.

Module 3 – AI Outreach Generation

This module helps sales teams create **AI-generated emails and outreach messages** based on the lead's information and industry. The aim is to make the outreach more relevant and personalized.

Module 4 – Lead Scoring & Recommendation Engine

This module evaluates leads and provides a **lead score** to help identify promising prospects. It also gives recommendations such as **when to follow up, which communication channel to use, and what type of content to send**.

Module 5 – Conversation Intelligence & CRM Integration

This module is planned to connect SalesGenie AI with **CRM systems** and help manage sales conversations. It will also support features such as **conversation or meeting summarization and data synchronization**.

Module 6 – Dashboard & Sales Analytics

This module provides a **centralized dashboard** where users can view important sales information and insights, such as the sales pipeline, conversion-related metrics, and follow-up activities.

Milestone – 1: QA Report

Testing Overview:

Milestone 1 was the first major testing checkpoint of the **SalesGenie AI** project. The main focus at this stage was to make sure that the initial features of the application were working properly before moving on to the next phase of development.

The milestone mainly covered **Lead Management, Prospect Database, and Lead Intelligence**. From the QA side, the focus was on checking the basic lead operations such as creating, viewing, updating, and deleting lead records. The backend APIs and database operations were also checked to make sure that the data was being processed, stored, and retrieved correctly.

Along with the basic lead management features, the initial **AI-based lead qualification functionality** was also reviewed using sample lead information. The purpose was to check whether the system was able to process the given information and provide a meaningful qualification result.

After the development work for these features was completed, testing was carried out to identify any functional or integration issues. Whenever an issue was found, it was discussed with the respective team member, corrected, and then tested again to make sure the correction worked properly.

Overall, Milestone 1 helped us verify that the basic building blocks of SalesGenie AI were working together correctly and provided a stable foundation for the features planned in the next milestones.

Testing Objectives:

The main goal of Milestone 1 testing was to make sure that the features developed so far were **working correctly and behaving as expected**. The testing focused on the following areas:

1. To make sure the **Lead Management** features were working according to the planned requirements.
2. To check whether leads could be **created, viewed, updated, and deleted** without any issues.
3. To verify that the **backend APIs** were receiving requests correctly and returning the expected responses.
4. To ensure that lead information was being **stored and retrieved correctly from the database**.
5. To check the **Lead Intelligence and qualification features** using sample lead information and review whether the generated results were meaningful.
6. To make sure the **frontend, backend, database, and AI components** were communicating properly with each other.

7. To identify any functional or integration issues before completing the **Milestone 1 checkpoint**.
8. To test the affected features again after corrections were made and confirm that the reported issues had been properly resolved.

Testing Scope:

The testing scope for Milestone 1 was mainly divided into the following areas.

Lead Management

Testing the basic operations performed on lead records:

- Creating a lead
- Viewing lead information
- Updating lead information
- Deleting a lead
- Searching/retrieving leads
- Validating lead inputs

Backend APIs

Testing the APIs responsible for Lead Management and checking:

- Request handling
- Response data
- HTTP status codes
- Input validation
- Error handling

Database

Checking whether:

- Lead records are stored correctly.
- Updated information is reflected correctly.
- Deleted records are handled properly.
- Retrieved information matches the stored data.

Lead Intelligence

Testing the initial AI-based functionality by:

- Providing sample lead/company information.

- Checking the generated qualification result.
- Reviewing the relevance of the output.
- Checking whether the result is displayed correctly.

Integration

Checking the overall communication between:

Frontend → Backend → Database / AI → Frontend

Testing Environment & Tools:

The following tools and technologies were used or planned for the Milestone 1 testing activities.

Area	Tool / Technology
API Testing	Postman
Backend	FastAPI
Database	PostgreSQL
ORM	SQLAlchemy
Frontend	Streamlit
AI Integration	Gemini / AI API
Automated Testing	PyTest
Version Control	Git
Manual Testing	Application UI / Browser

Testing Approach:

For Milestone 1, we used a combination of **manual testing and API testing**. Postman was used to directly test the backend APIs and check whether they were responding correctly. At the same time, the application interface was checked manually to make sure that the data and results received from the backend were being displayed properly to the user.

This approach helped us test the application from both the **backend side and the user's side**, making it easier to identify issues in individual features as well as in the overall workflow.

Test Cases:

Test ID	Module	Test Scenario	Expected Result	Actual Result	Status
TC-01	Lead Management	Create a new lead	Lead should be created successfully	Lead was created successfully	Pass
TC-02	Lead Management	Retrieve an existing lead	Correct lead details should be returned	Correct lead details were displayed	Pass
TC-03	Lead Management	Update lead information	Updated details should be saved correctly	Updated details were saved correctly	Pass
TC-04	Lead Management	Delete a lead	Selected lead should be deleted successfully	Lead was deleted successfully	Pass
TC-05	Lead Management	Submit invalid lead information	System should reject invalid input	Invalid input was rejected	Pass
TC-06	Backend API	Verify API response	API should return the expected response/status	Expected response was received	Pass
TC-07	Database	Verify stored lead data	Stored data should match submitted data	Stored data matched the submitted data	Pass
TC-08	Database	Retrieve updated lead	Latest information should be returned	Updated information was retrieved correctly	Pass
TC-09	Lead Intelligence	Generate qualification result	AI should provide a relevant qualification result	Qualification result was generated	Pass
TC-10	Integration	Submit lead through UI	Data should reach backend and be processed correctly	Data was successfully processed	Pass

Test Execution Metrics:

Test execution metrics provide a quick overview of how many test cases were performed, how many passed or failed during the initial execution, and how the identified issues were handled.

For Milestone 1, the testing was carried out across **Lead Management, Backend APIs, Database Operations, Lead Intelligence, and Integration.**

Metric	Result
Total Test Cases	10
Test Cases Executed	10
Passed – Initial Execution	8
Failed – Initial Execution	2
Blocked	0
Initial Pass Percentage	80%
Defects Identified	2
Defects Resolved	2
Defects Retested	2
Final Passed After Retesting	10
ss	100%

Pass Percentage – Initial Execution

The initial pass percentage was calculated using:

Pass Percentage = (Passed Test Cases ÷ Executed Test Cases) × 100

= (8 ÷ 10) × 100

= 80%

The two failed test cases resulted in defects that were reported to the respective team members. After the issues were corrected, both test cases were executed again.

Final Testing Result

After the identified issues were fixed and retested, all **10 test cases passed successfully**.

Final Pass Percentage = (10 ÷ 10) × 100 = 100%

Defect Summary:

During Milestone 1 testing, **two defects** were identified in the Lead Management APIs. Both were related to input validation and were later resolved.

Defect ID	Module	Issue	Severity	Status
BUG-01	Lead Management	Required field was accepted as empty while creating a lead.	Medium	Resolved
BUG-02	Lead Management	Invalid input was accepted while updating a lead.	Medium	Resolved

Severity

- **Critical:** Major functionality is completely blocked.
- **High:** Important functionality is seriously affected.
- **Medium:** A specific feature is affected but the application can still be used.
- **Low:** Minor UI or usability issue.

Both defects were classified as **Medium** because they affected data validation but did not stop the application from functioning.

Defect Resolution & Retesting:

The identified defects were shared with the respective team member and corrected. After the fixes were completed, the same test scenarios were executed again to confirm that the issues had been resolved.

Defect ID	Issue	Retest Result	Final Status
BUG-01	Required field validation	Passed	Closed
BUG-02	Update input validation	Passed	Closed

Summary

Both identified defects were **fixed and successfully verified through retesting**. After retesting, the affected test cases passed successfully, allowing the Milestone 1 testing to proceed toward completion.

This is much cleaner for the report while still showing the complete QA process:

2 defects → 2 fixes → 2 retests → 2 passed → 2 closed.

Integration Testing:

Integration testing was done to make sure that the different parts of **SalesGenie AI** were **working together properly**. Instead of testing each component separately, we checked how the data moved through the complete application.

The main flow checked was:

Frontend → FastAPI Backend → Database / AI → Backend → Frontend

Areas Checked

- Checked whether the **frontend was sending the correct information** to the backend.
- Checked whether the **backend was receiving and processing the requests** correctly.
- Verified that lead information was being **stored and retrieved properly from PostgreSQL**.
- Checked whether the **AI component was receiving the required lead information**.
- Verified that the **AI-generated result was returned correctly** to the backend.
- Finally, checked whether the processed information and results were **displayed correctly on the frontend**

AI / Lead Qualification Validation:

Since AI is an important part of **SalesGenie AI**, the Lead Intelligence and qualification functionality was also included in the Milestone 1 testing.

Sample lead and company information was provided to the system, and the generated qualification results were reviewed to make sure they were relevant to the information provided.

Areas Checked

- Whether the system accepted the required lead information correctly.
- Whether a qualification result was generated successfully.
- Whether the generated result was relevant to the given lead information.
- Whether similar inputs produced reasonably consistent results.
- Whether the result was displayed correctly in the application.
- Whether the AI functionality was properly connected with the rest of the application.

QA Observation

The testing focused not only on whether the AI produced a result, but also on whether the result **made sense in relation to the lead and company information provided**. The generated qualification results were reviewed for relevance and consistency before considering the feature ready for the milestone.

Overall QA Results:

The following table summarizes the main areas covered during **Milestone 1 testing**:

Testing Area	Result
Lead CRUD Operations	Passed
Backend API Testing	Passed
Database Validation	Passed
Lead Intelligence	Passed
Frontend Validation	Passed
Integration Testing	Passed
Defect Retesting	Passed

Overall QA Summary

Milestone 1 testing covered the main **Lead Management and Lead Intelligence** features of SalesGenie AI. The testing included CRUD operations, API and database checks, AI-based qualification validation, frontend verification, and integration testing.

Two issues were identified during the initial testing and were later corrected and successfully retested. After the fixes, the affected test cases passed successfully.

Overall, the Milestone 1 features were **tested, reviewed, and found ready to move forward to the next stage of development.**

Conclusion:

Milestone 1 testing was completed to make sure that the initial features of **SalesGenie AI** were working properly and communicating correctly with each other. The QA activities covered **Lead Management, backend APIs, database operations, Lead Intelligence, frontend functionality, and integration testing.**

During testing, a few issues were identified and shared with the respective team members. After the required corrections were made, the affected features were tested again and the fixes were successfully verified.

Overall, the Milestone 1 features were **tested and reviewed successfully**, providing a stable foundation for moving forward with the next phase of the project.